

## Chapter 2 – Review of Linear Regression

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### 1 Introduction

1. We see that in modelling various time series models, similar techniques to linear regression are used.
2. Thus we review techniques of simple linear regression before looking at time series.
3. In simple linear regression, we have the predictor or independent variable  $x_i$  and the response variable  $y_i$ .
4. The problem typically is that there is a linear relationship between  $y_i$  and  $x_i$ , for example, weight as a linear function of height of adults. So, given the height of a typical adult, can we predict the person's weight? So here height is the independent variable and weight is the response variable.
5. We typically would have already observed the paired data weight and height  $(y_i, x_i)$  of a sample of  $n$  adults, so we use the observe data to find a relationship

$$y_i = c + ax_i + z_i, \text{ for } i = 1, 2, \dots, n,$$

where  $z_i$  is the residual error of the  $i$ th observation.

6. The intercept  $c$  and the gradient  $a$  have to be estimated from the observed data.

7. In time series, there isn't a response variable, but for example, we may want to find out whether  $x_k$  is related to the observation  $x_{k-1}$ , one period before it. If we believe that there is a linear relationship, then we could try to fit a model

$$x_k = c + ax_{k-1} + z_k.$$

Here, the observations are used to estimate  $a$  instead.

8. We see that the two equations look similar.
9. In fact, many common time series models can be broken down into more linear models so that linear regression techniques can be used.
10. We see later that the model in item 1.7 above is called an *Autoregressive model* of order 1 or *AR(1)* model.
11. It is called autoregressive model because the variable  $x_k$  is regressed on preceding observations.

## 2 Simple Linear Regression

1. We look at the simplest form of linear regression where there is only 1 independent variable such as the model in item 1.5, that is,

$$y_i = c + ax_i + z_i, \text{ for } i = 1, 2, \dots, n,$$

where  $z_i$  is the residual error of the  $i$ th observation.

2. Assuming that the predictor variable and response of each observation  $(x_i, y_i)$  are independent, then the residual errors  $z_i$  can also be assumed to be independent.
3. Since we are dealing with a large number of observations, we can assume that each  $z_i$  are independently and identically distributed as normal with zero mean and constant spread (standard deviation)  $\sigma$ . That is,

$$z_i \stackrel{i.i.d.}{\sim} N(0, \sigma).$$

4. The traditional method for estimating the intercept parameter  $c$  and the gradient parameter  $a$  is to apply the *Ordinary Least Squares* method (OLS).
5. First we note that for each pair of predictor-response from each observation  $(x_i, y_i)$ , the residual error is

$$z_i = y_i - c - ax_i,$$

which can be either positive or negative.

6. The squared residual error is

$$z_i^2 = (y_i - c - ax_i)^2.$$

7. From the  $n$  observations, the sum of square error is

$$SSE = \sum_{i=1}^n z_i^2 = \sum_{i=1}^n (y_i - c - ax_i)^2.$$

8. Let  $\hat{c}$  and  $\hat{a}$  be the solutions to  $c$  and  $a$  that minimises the SSE in item 2.7, then through calculus and algebra, it can be shown that

$$\hat{a} = \frac{\frac{\sum_{i=1}^n x_i y_i}{n} - \bar{x}\bar{y}}{\frac{\sum_{i=1}^n x_i^2}{n} - (\bar{x})^2},$$

and

$$\hat{c} = \bar{y} - \hat{a}\bar{x},$$

where  $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$  and  $\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$  are the sample means of the  $x_i$  and the  $y_i$  respectively.

9. **Example 1.** In **R**, there is this data set `cars` where the speed in mph and stopping distance in feet of 50 cars are observed. First we type

```
> data(cars)
```

so that **R** knows that we are using the `cars` data set.

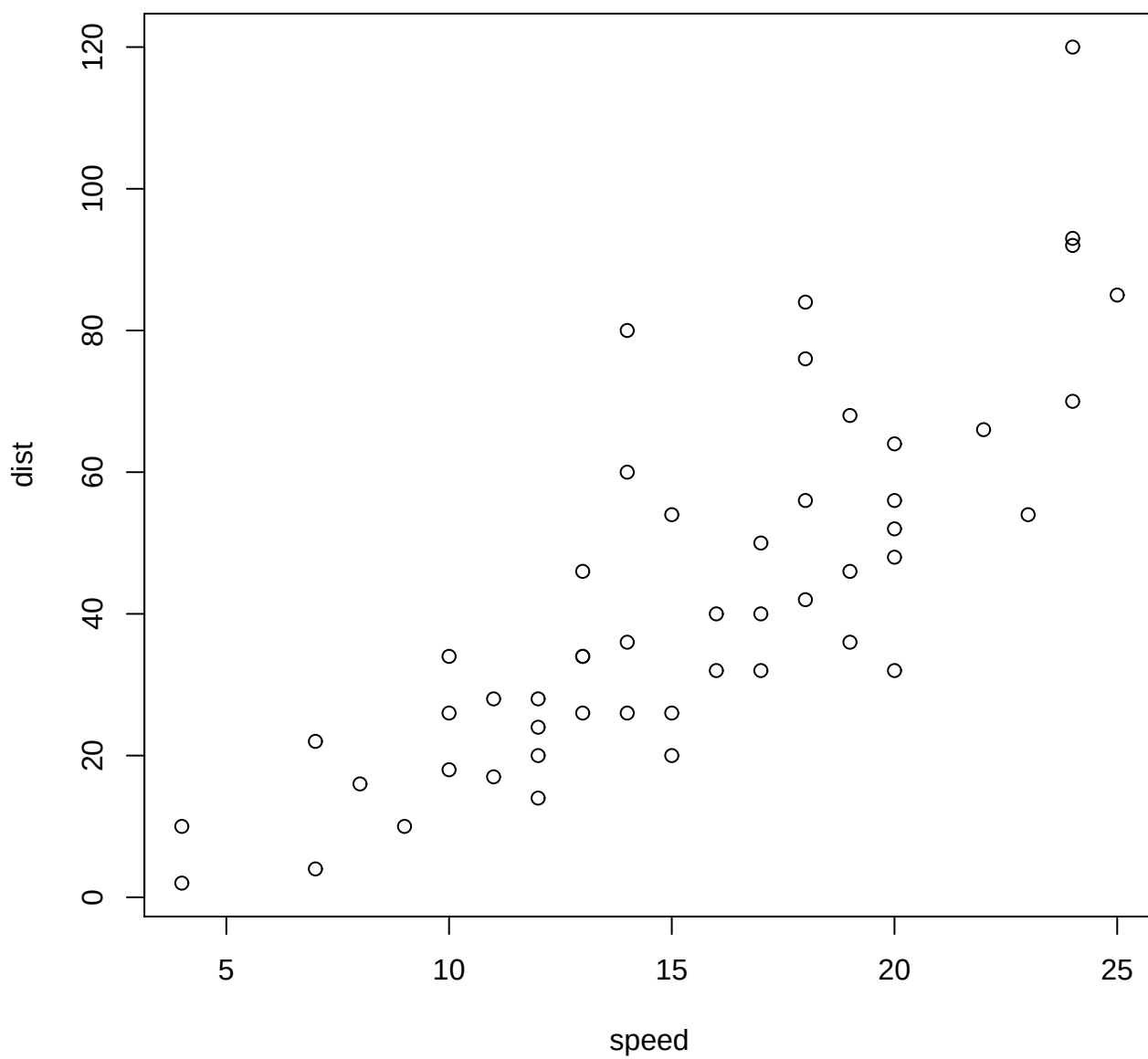
In this case, each entry has two attributes, where they are already called `speed` and `dist` in the data, so to call up any of them alone, we use `cars$speed` and `cars$dist`. We can do a `summary(cars)` to get a summary of both variables or do them separately.

```
> summary(cars$speed)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   4.0   12.0   15.0   15.4   19.0   25.0

> summary(cars$dist)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 2.00  26.00  36.00  42.98  56.00 120.00
```

We do a `plot(cars)` or `plot(cars$speed, cars$dist)` to see if there is a linear relationship. The command `plot(cars)` work in this case because the data only has two fields and by default `speed` is plotted on the  $x$ -axis and `dist` is plotted on the  $y$ -axis.

Figure 1: Plot of Stopping Distance against Speed



From the plot in Figure 1, it certainly looks like there is a linear relationship. Let us now estimate the parameters and see how good the fit is.

```
> fit <- lm(cars$dist ~ cars$speed)
> fit
```

```
Call:
lm(formula = cars$dist ~ cars$speed)
```

```
Coefficients:
(Intercept)  cars$speed
   -17.579      3.932
```

This tells us that  $\hat{c} = -17.579$  and  $\hat{a} = 3.932$ . You can also try using the formulas item 2.8 to calculate using a calculator.

```
> summary(fit)
```

```
Call:
lm(formula = cars$dist ~ cars$speed)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-29.069  -9.525  -2.272   9.215  43.201
```

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -17.5791     6.7584  -2.601   0.0123 *
cars$speed   3.9324     0.4155   9.464 1.49e-12 ***
---
Signif. codes:  0 *** 0.001 ** 0.01 * 0.05 . 0.1 1
```

Residual standard error: 15.38 on 48 degrees of freedom  
Multiple R-squared: 0.6511, Adjusted R-squared: 0.6438  
F-statistic: 89.57 on 1 and 48 DF, p-value: 1.49e-12

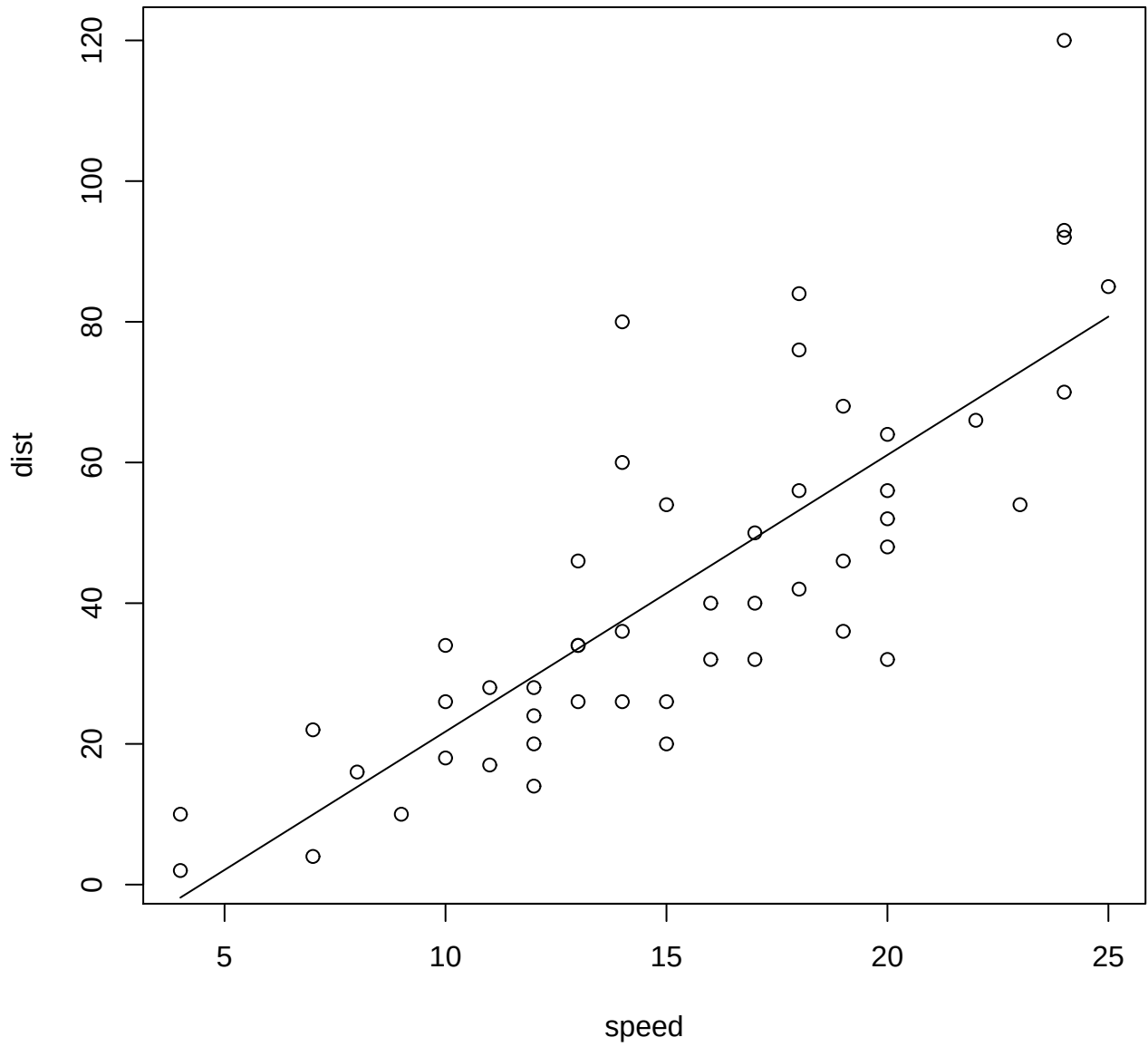
This tells us that the estimates of both the intercept and the gradient are significantly different from zero (see the  $\Pr(> |t|)$  column). Both probabilities are less than 0.05 if we are talking about a 5% significance level.

Now we plot the data again with the fitted line to see how it looks like.

```
> plot(cars)
> lines(cars$speed, fit$fitted.values)
```

The plot with the fitted line is shown in Figure 2.

Figure 2: Plot of Cars Data with Fitted Line





Now we need to check whether the fitted residuals are normally distributed and whether there are any trends or outliers in the residuals.

```
> qqnorm(fit$residuals)
```

produces the Q-Q plot in Figure 3.

We also do the residual plot and add a horizontal line through zero.

Now we check the standard deviation of the fitted residuals.

```
> sd(fit$residuals)
[1] 15.22184
```

Thus 2 standard deviations is approximately 30.44. We do the residuals plot and check whether there are any outliers more than 2 standard deviations away on either side of zero.

```
> plot(fit$fitted,fit$residual)
> abline(0,0)
> abline(30.44,0)
> abline(-30.44,0)
```

The plot is shown in Figure 4 and there are 3 positive residuals greater than 2 standard deviations.

Figure 3: Normal Q-Q Plot of the Residuals

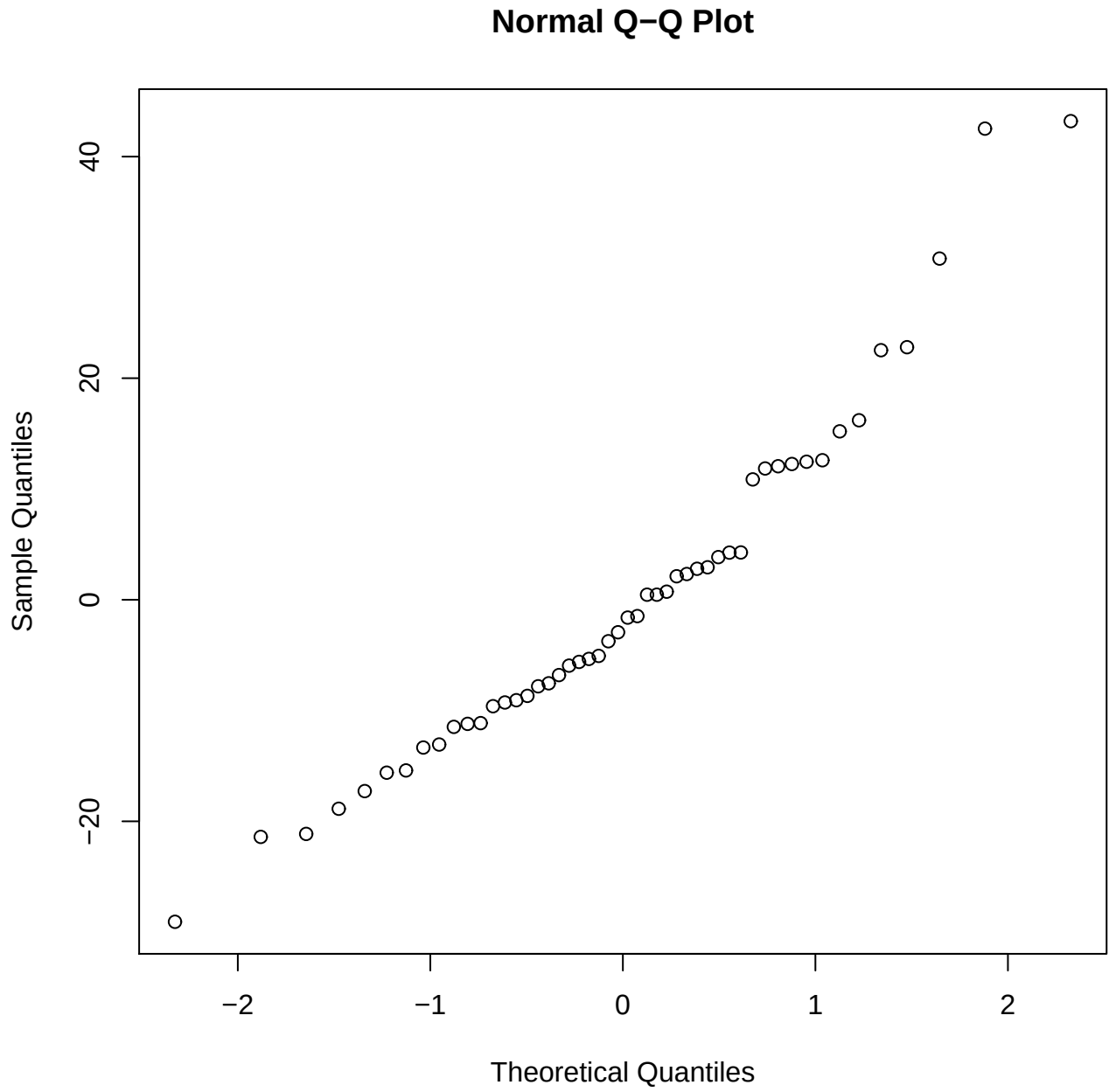
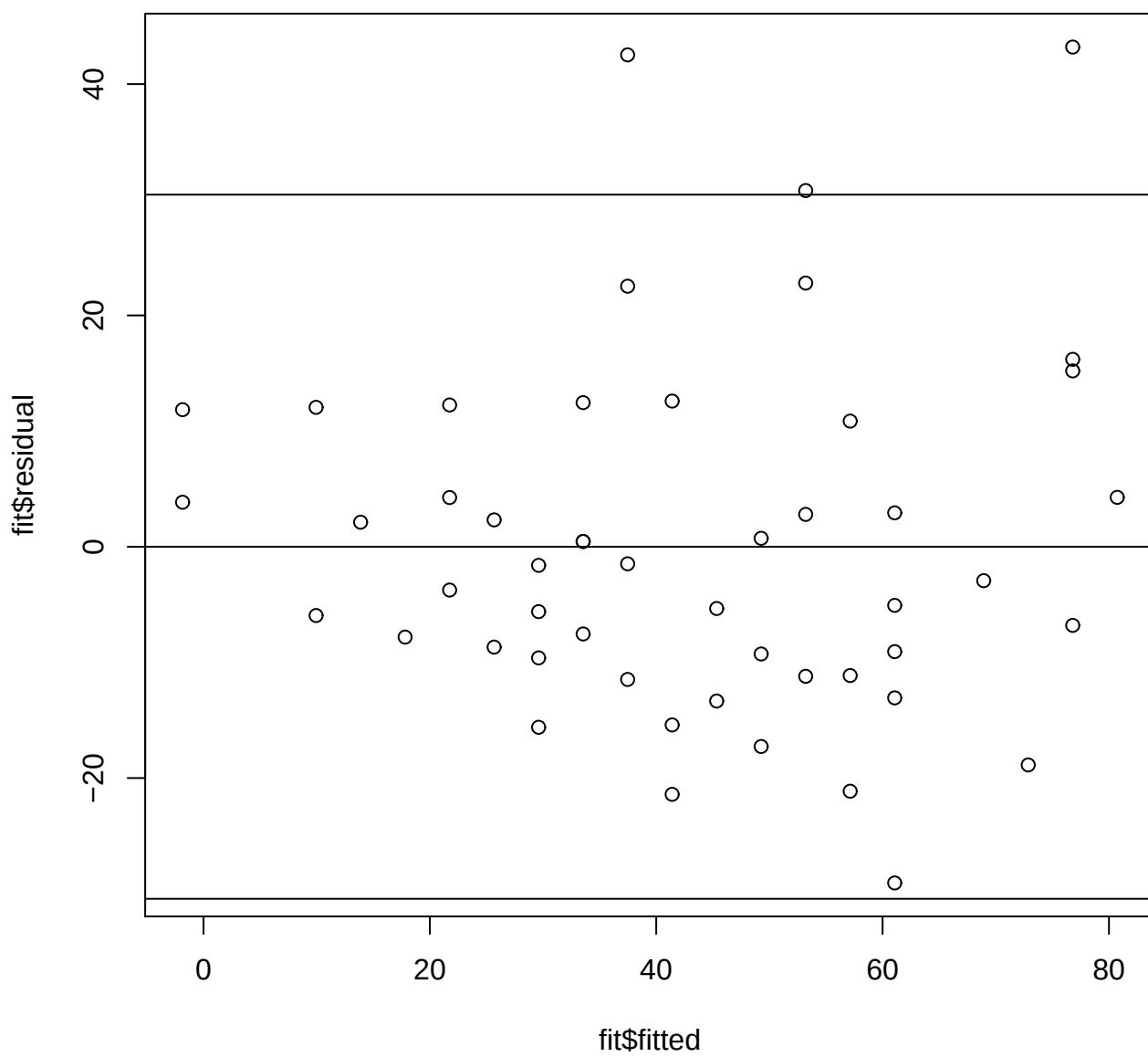


Figure 4: Residual Plot



Thus we have to identify those observations that contribute to those 3 significantly large residuals, remove them from the data and do the fit all over again.

We do

```
> fit$residuals
```

| 1          | 2          | 3          | 4          | 5          | 6          | 7         |
|------------|------------|------------|------------|------------|------------|-----------|
| 3.849460   | 11.849460  | -5.947766  | 12.052234  | 2.119825   | -7.812584  | -3.744993 |
| 8          | 9          | 10         | 11         | 12         | 13         | 14        |
| 4.255007   | 12.255007  | -8.677401  | 2.322599   | -15.609810 | -9.609810  | -5.609810 |
| 15         | 16         | 17         | 18         | 19         | 20         | 21        |
| -1.609810  | -7.542219  | 0.457781   | 0.457781   | 12.457781  | -11.474628 | -1.474628 |
| 22         | 23         | 24         | 25         | 26         | 27         | 28        |
| 22.525372  | 42.525372  | -21.407036 | -15.407036 | 12.592964  | -13.339445 | -5.339445 |
| 29         | 30         | 31         | 32         | 33         | 34         | 35        |
| -17.271854 | -9.271854  | 0.728146   | -11.204263 | 2.795737   | 22.795737  | 30.795737 |
| 36         | 37         | 38         | 39         | 40         | 41         | 42        |
| -21.136672 | -11.136672 | 10.863328  | -29.069080 | -13.069080 | -9.069080  | -5.069080 |
| 43         | 44         | 45         | 46         | 47         | 48         | 49        |
| 2.930920   | -2.933898  | -18.866307 | -6.798715  | 15.201285  | 16.201285  | 43.201285 |
| 50         |            |            |            |            |            |           |
| 4.268876   |            |            |            |            |            |           |

to see which observations contribute to the 3 significantly large residuals. These are observations 23, 35 and 49. So we can remove them from cars and create a new data set called cars2

```
> cars2 <- cars[-c(23, 35, 49), ]
```

You can now check that cars2 has those observation removed and it is left as a tutorial exercise to redo all the fitting with cars 2.

### 3 Multiple Linear Regression

1. We now look at *Multiple Linear Regression*.
2. Basically in multiple linear regression, there are multiple independent variables (predictor variable).
3. There could also be multi-dimensional responses but we restrict ourselves to 1-dimensional responses.
4. We assume that there are  $n$  observations  $(x_{i1}, x_{i2}, \dots, x_{im}, y_i)$  where  $x_{ij}$  is the  $j$ th predictor variable of the  $i$ th observation and  $y_i$  is the response.
5. The model is

$$y_i = c + \sum_{j=1}^m a_j x_{ij} + z_i, \text{ for } i = 1, 2, \dots, n,$$

where  $z_i$  is the residual error of the  $i$ th observation.

6. The residual error is

$$z_i = y_i - c - \sum_{j=1}^m a_j x_{ij},$$

which can be either positive or negative.

7. The squared residual error is

$$z_i^2 = \left( y_i - c - \sum_{j=1}^m a_j x_{ij} \right)^2.$$

8. From the  $n$  observations, the sum of square error is

$$SSE = \sum_{i=1}^n z_i^2 = \sum_{i=1}^n \left( y_i - c - \sum_{j=1}^m a_j x_{ij} \right)^2.$$

9. The least squares method seeks to find estimates  $\hat{c}$  and  $\hat{a}_j$  that minimises the SSE given the observations.
10. Closed-form formulas are available for  $\hat{c}$  and  $\hat{a}_j$  if the problem is rewritten in a vector-matrix notation. However, we will not be doing this. We will simply use **R** to find the solutions.
11. **Example 2.** We use the trees data in **R** .

```
> data(trees)
```

This data set provides measurements of the girth, height and volume of timber in 31 felled black cherry trees. Note that girth is the diameter of the tree (in inches) measured at 4 ft 6 in above the ground.

Let  $V$  denote volume,  $H$  denote height and  $G$  denote girth. It is known that there is a relationship

$$V = kH^{a_1}G^{a_2},$$

where  $k$  is some constant. By using log scale, we obtain

$$\log V = \log k + a_1 \log H + a_2 \log G.$$

So we try to fit the model

$$\log V = c + a_1 \log H + a_2 \log G,$$

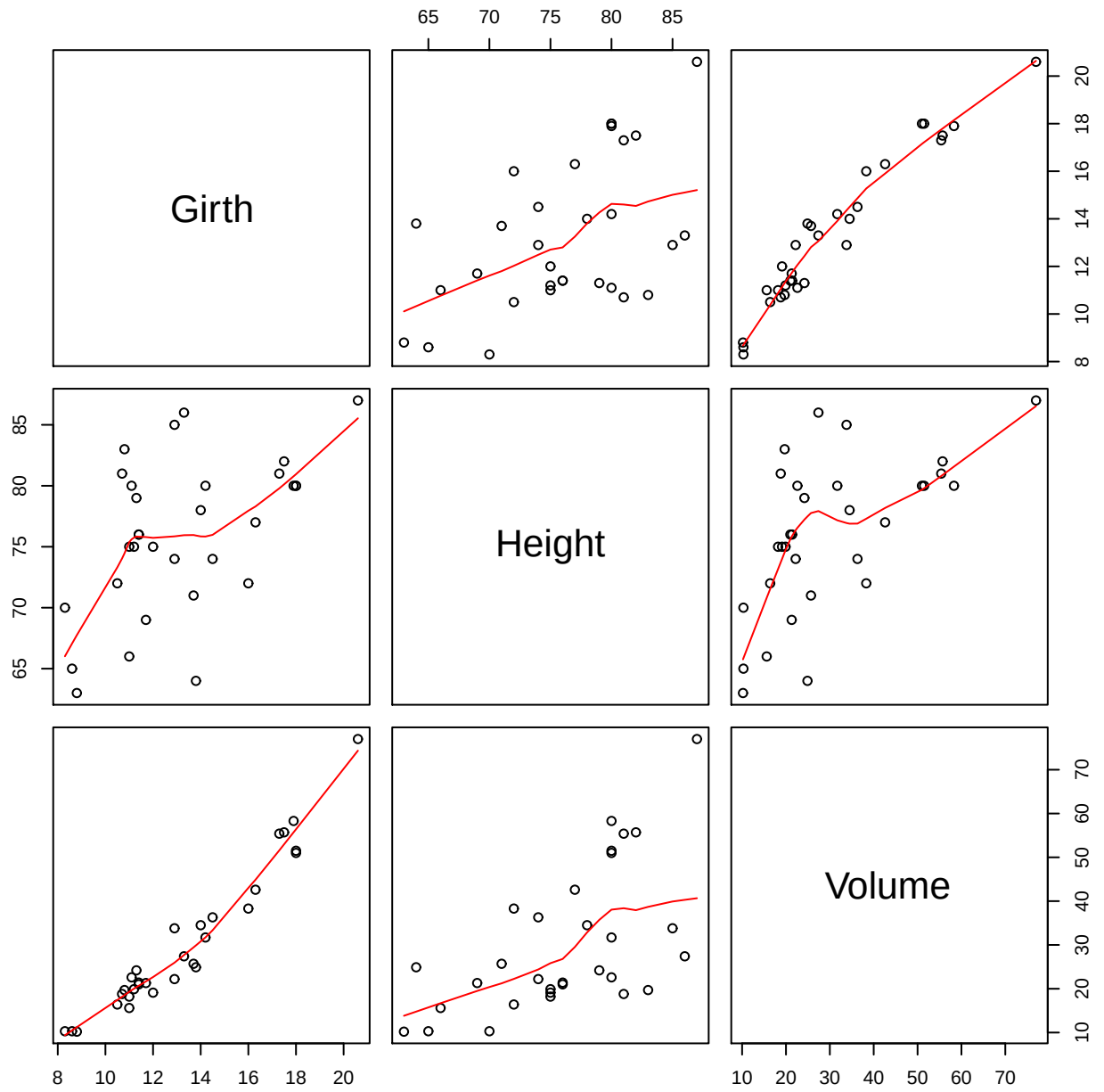
where  $c = \log k$ .

We do

```
> pairs(trees, panel = panel.smooth)
```

to get plots between pairs of each variable. The `panel = panel.smooth` option gives us the red curve to help us check whether there is a linear relationship between the pairs of variables. This gives us the plot in Figure 5.

Figure 5: Trees Pairs



We first plot Volume versus Girth under log scale to see if there is a linear relationship.

```
> plot(Volume ~ Girth, data = trees, log = "xy")
```

In Figure 6, we see that there is a linear relationship under the log scale.

We can also check log Volume versus log Girth given different intervals of Height just to see if the linear relationship holds over different ranges of Height.

```
> coplot(log(Volume) ~ log(Girth) | Height, data = trees, panel=panel.smooth)
```

In Figure 7, it appears that the relationship is also linear over different ranges of Height.



Figure 6: Volume versus Girth under Log Scale

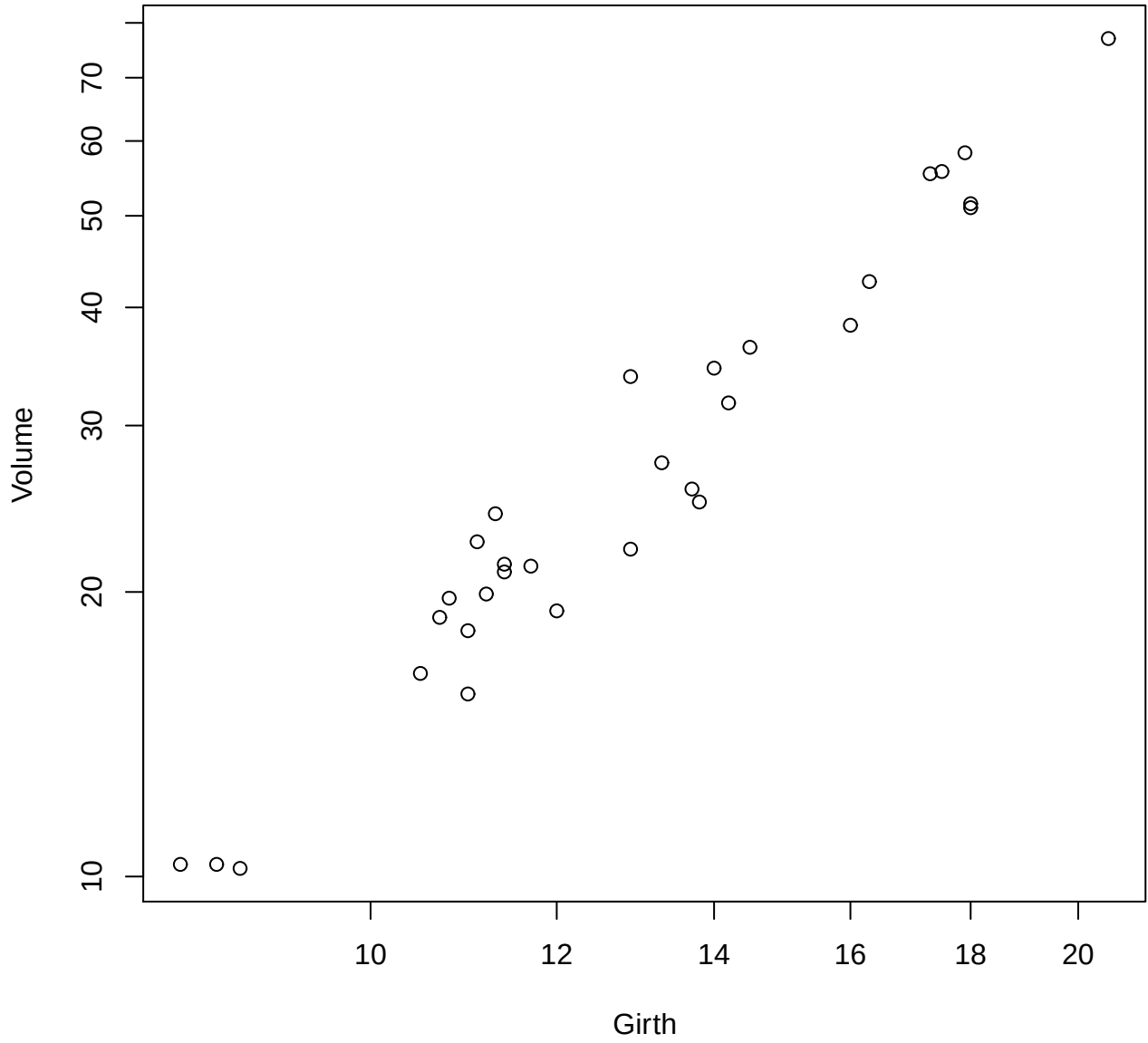
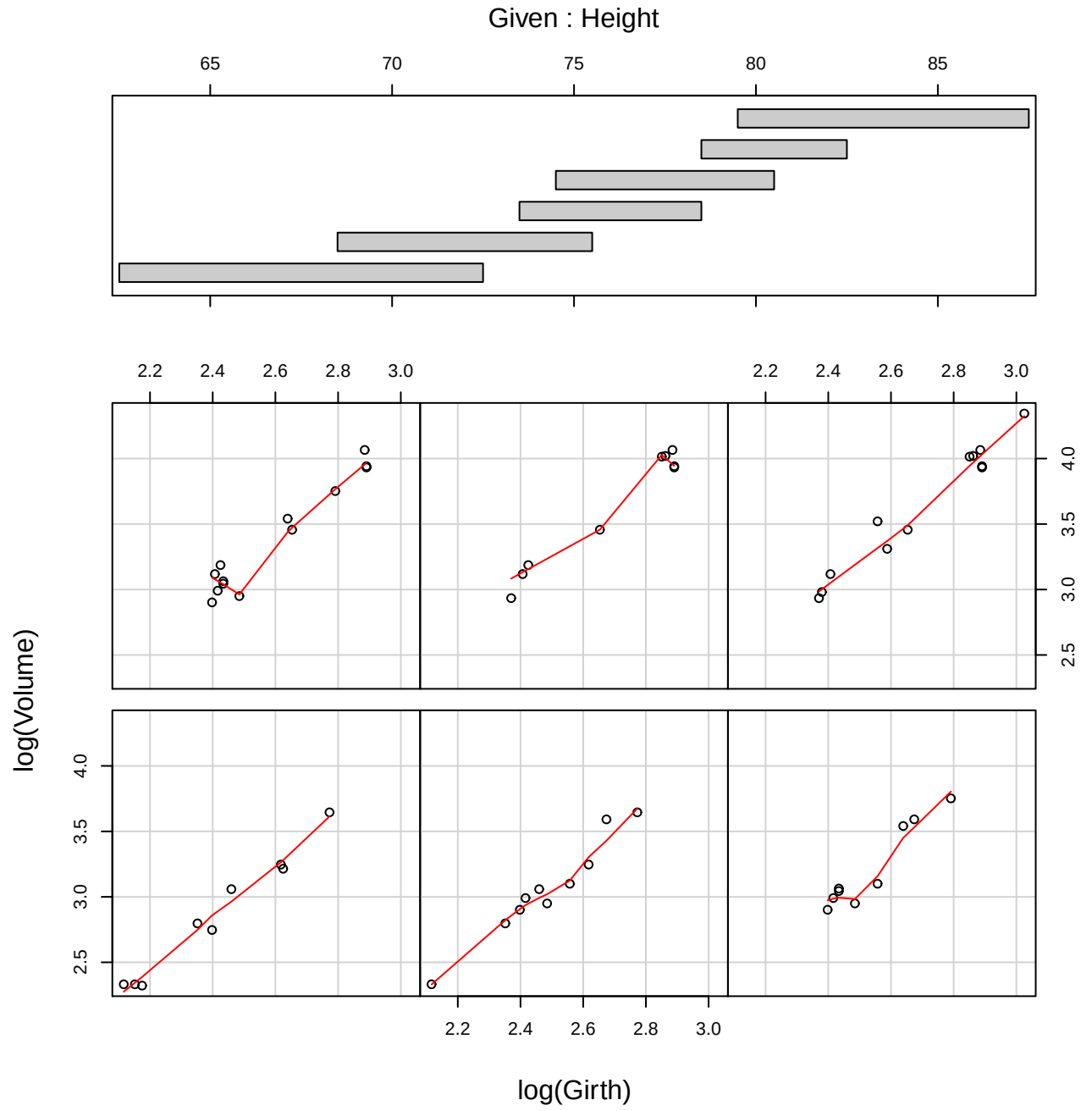


Figure 7: Coplot of Log Volume versus Log Girth given Height



Now we fit the model  $\log V = c + a_1 \log H + a_2 \log G$ .

```
> fit <- lm(log(Volume) ~ log(Girth) + log(Height), data=trees)
> fit

Call:
lm(formula = log(Volume) ~ log(Girth) + log(Height), data = trees)

Coefficients:
(Intercept)    log(Girth)    log(Height)
      -6.632         1.983         1.117

> summary(fit)

Call:
lm(formula = log(Volume) ~ log(Girth) + log(Height), data = trees)

Residuals:
      Min       1Q   Median       3Q      Max
-0.168561 -0.048488  0.002431  0.063637  0.129223

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  -6.63162    0.79979  -8.292 5.06e-09 ***
log(Girth)    1.98265    0.07501  26.432 < 2e-16 ***
log(Height)   1.11712    0.20444   5.464 7.81e-06 ***
---
Signif. codes:  0 *** 0.001 ** 0.01 * 0.05 . 0.1 1

Residual standard error: 0.08139 on 28 degrees of freedom
Multiple R-squared:  0.9777,    Adjusted R-squared:  0.9761
F-statistic: 613.2 on 2 and 28 DF,  p-value: < 2.2e-16
```

From the results, all parameters are significant and the estimates are  $\hat{c} = -6.63162$ ,  $\hat{a}_1 = 1.11712$  and  $\hat{a}_2 = 1.98265$  based on the data.

We can check the normality of the residuals using `qqnorm(fit$residuals)`. The residuals appear to be normally distributed as in Figure 8. One standard deviation of the residuals is about 0.079, so in the plot of the residuals against fitted values in Figure 9, it appears that only 2 residuals are marginally less than 2 standard deviations. So we can leave the results as they are, otherwise we have to go through the procedure of removing those observations and doing the analysis all over again.

It can be noted that the theoretical relationship is

$$V = kHG^2,$$

so our estimates based on the data set are not too far off the theoretical values of the parameters.

Figure 8: Normal Q–Q Plot of Residuals

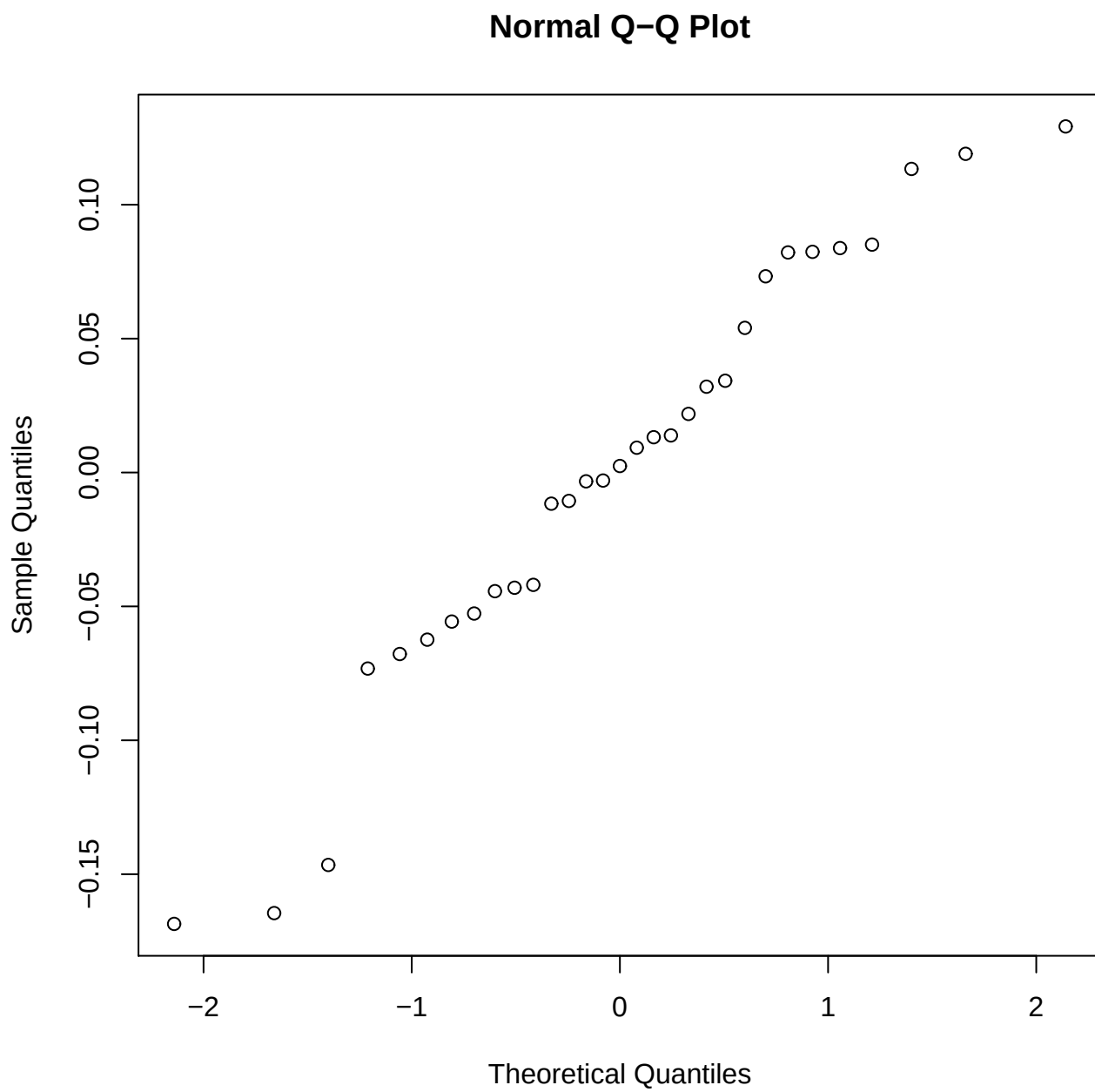
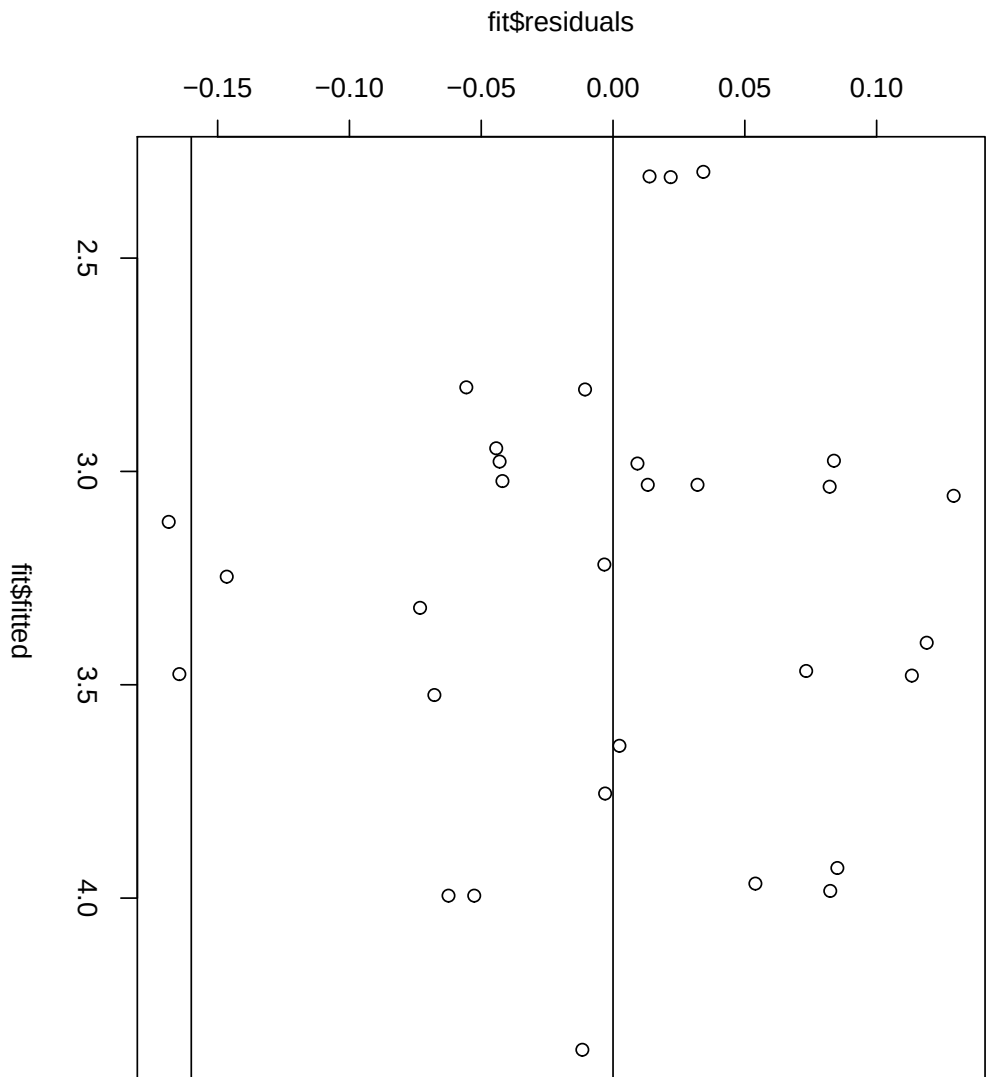


Figure 9: Residual Plot



## Tutorial 2 Questions.

1. Looking at the cars data, remove those 3 observations that have outlier residuals and do the analysis all over again, that is, do the model fitting and plots, and re-check the residuals.
2. Looking at the trees data, suppose we had missed out the height measurements but only took the volume and girth measurements, do a plot to show that this looks like there is a linear relationship, fit the linear model between volume and girth, do the relevant plots and check the residuals.
3. Consider the rock data in **R** . Check whether perimeter is a good explanatory variable for area.
4. Consider the mtcars data in **R** . Check whether some of the continuous variables such as displacement, horse power, weight and real axle ratio are good explanatory variables for fuel consumption.